Thermal Control on a CFRP Sandwich Structure

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Mathematical Terminology

E	Error for the Proportional-Integral Control Law
L_{xp}	Panel length along the x-axis
L_{zp}	Panel length along the z-axis
$L_{x\text{EBOX}}$	EBOX length along the x-axis
$L_{z\text{EBOX}}$	EBOX length along the z-axis
$L_{x\text{HT}}$	Heater length along the x-axis
$L_{z\text{HT}}$	Heater length along the z-axis
$h_{\text{P-EBOX}}$	Contact conductance between the panel and the EBOX
h_{ef}	Effective contact conductance
c_{EBOX}	Thermal capacity of the EBOX
W_{EBOX}	Heat dissipation of the EBOX
W_{HT}	Heat dissipation of the heater
ρ_{CFRP}	Density of the CFRP material
ρ_{HC}	Density of the Honeycomb material
$\bar{\rho}$	Average density
c_{CFRP}	Specific heat capacity of the CFRP material
c_{HC}	Specific heat capacity of the Honeycomb material
c_{ef}	Effective specific heat capacity
k_{CFRP}	Thermal conductivity of the CFRP material
k_{HC}	Thermal conductivity of the Honeycomb material
$k_{\text{ef, along}}$	Effective thermal conductivity
$T_{\text{N-ON EBOX}}$	Nominal operating temperature of the EBOX
$T_{\text{S-OFF EBOX}}$	Shutdown and survival mode temperature of the EBOX
$\dot{q}_v$	Heat dissipation per unit volume
$L_{y\text{HC}}$	Thickness of the Honeycomb panel
$L_{y\text{CFRP}}$	Thickness of the CFRP material
t	Time
δ_t	Total structural thickness
T	Temperature
T_R	Satellite temperature
T_d	Desired (target) temperature for Proportional-Integral Control
K_p	Proportional gain
K_i	Integral gain
α	Thermal diffusivity
Δt	Time step for the numerical scheme
Δx	Spatial step for the numerical scheme
Fo	Fourier number for stability
N	Number of nodes belonging to the panel (excluding the EBOX node)
N_e	Panel nodes in contact with the EBOX
N_p	First panel node in contact with the EBOX
N_q	Last panel node in contact with the EBOX
U	Total Proportional-Integral Control variable
U_p	Proportional control variable
U_I	Integral control variable

1. Introduction

The purpose of this document is to explain the physical and mathematical fundamentals behind the Python code developed to simulate heat transfer and thermal control in a sandwich panel, as typically found in the space industry. An EBOX is installed on the panel, together with a heater designed to keep the electronic board within its survival temperature range at all times.

The panel considered in this study consists of an aluminium honeycomb core with a hexagonal cell structure and outer CFRP layers. During its operation in space, the panel is exposed to significant temperature variations. The main objective of this study is to investigate and understand how the temperature distribution across the panel changes under different operating conditions and heat dissipation configurations. The structure considered in this study is shown in Figure (1).

Through an analytical and numerical approach, several situations will be studied, ranging from the analysis of the steady-state temperature distribution across the panel to the development of thermal control strategies. These strategies aim to keep the temperatures of critical components, such as the electronic box, within specific limits. In addition, the effect of thermal radiation will be explored, and suitable control systems will be designed to ensure that the temperature requirements are met throughout all phases of satellite operation.

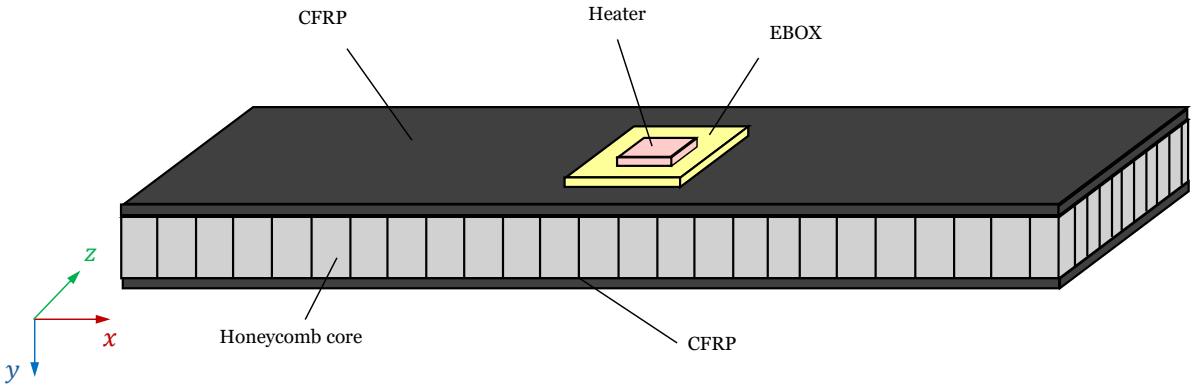


Figure 1. Isometric view of the panel structure considered in this study.

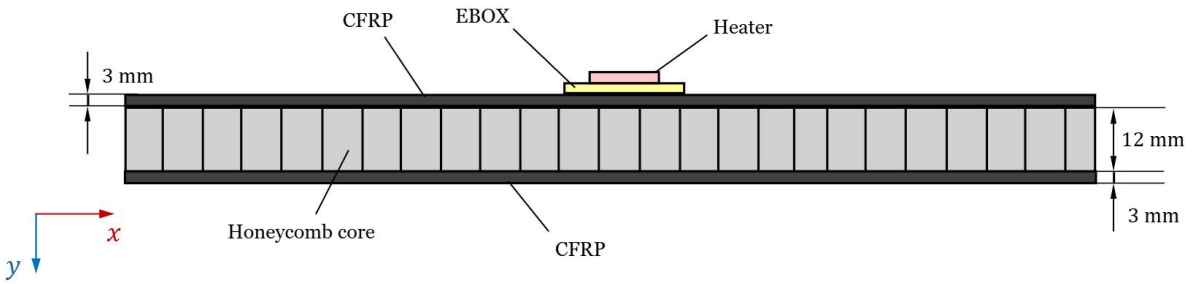


Figure 2. Front view of the panel structure considered in this study.

The required properties of the different parts of the structure, including both thermo-physical [1] and dimensional properties, are listed in Tables (1) and (2).

Table 1. Dimensional properties of the structure.

Dimension	Value
L_{xp} [mm]	600
L_{zp} [mm]	400
$L_{y_{\mathrm{CFRP}}}$ [mm]	3
$L_{y_{\mathrm{HC}}}$ [mm]	12
$L_{x_{\mathrm{EBOX}}}$ [mm]	200
$L_{z_{\mathrm{EBOX}}}$ [mm]	200
$L_{x_{\mathrm{HT}}}$ [mm]	100
$L_{z_{\mathrm{HT}}}$ [mm]	100

Table 2. Thermal and physical properties of the structure.

Property	Value
$h_{\text{P-EBOX}}$ [W/(m^{2}K)]	200
c_{EBOX} [J/K]	1900
W_{EBOX} [W]	50
W_{HT} [W]	12
ρ_{CFRP} [kg/m^3]	1650
ρ_{HC} [kg/m^3]	40
c_{CFRP} [J/(kg K)]	930
c_{HC} [J/(kg K)]	890
k_{CFRP} [W/(m K)]	130
k_{HC} [W/(m K)]	0.7
$T_{\text{N-ON EBOX}}$ [°C]	$10 - 60$
$T_{\text{S-OFF EBOX}}$ [°C]	$5 - 70$

The panel is attached to the satellite structure along its two shorter sides. For the cold case considered in this study, the satellite structure is assumed to be at a temperature of 0°C. The two longer sides of the panel are considered insulated.

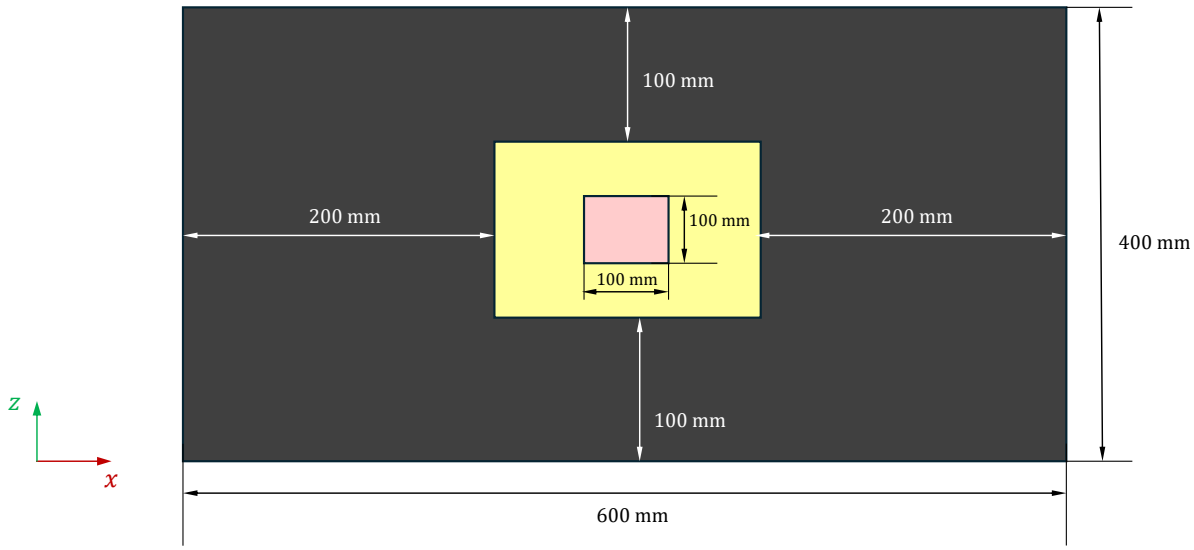


Figure 3. Top view of the panel structure considered in this study.

2. Mathematical Model

This section presents the development of the mathematical formulation required to solve the problem under the different scenarios and study cases. Starting from a simplified approach and gradually moving towards a more detailed and realistic representation of the model, the aim is to build a complete understanding of the thermal system under consideration.

2.1. Heat Transfer

The heat equation is a fundamental tool for studying heat transfer processes in different physical systems. Equation (1), derived from the law of conservation of energy, describes how the temperature of a medium changes with time and spatial position. It takes into account the thermal properties of the material, as well as any heat sources or sinks present in the medium, in a one-dimensional environment.

$$\rho c \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2} + \dot{q_v}, \tag{1}$$

where ρ represents the material density, c its specific heat capacity, T the temperature, t the time variable, k the thermal conductivity, x the spatial variable, and $\dot{q_v}$ represents any heat source or sink per unit volume that may be present in the medium. This term can represent, for example, heat generated by internal processes within the material or heat lost to the surroundings.

The term $\partial T/\partial t$ represents the rate of change of temperature with time at a given point in the medium, while $\partial^2 T/\partial x^2$ represents the spatial variation of temperature at that same point. Together, these two contributions describe how temperature changes over time and space.

2.2. Analytical Solution

First, it is assumed that the electronic box and the heater extend across the entire shorter side of the panel, so that a one-dimensional temperature distribution can be considered. In addition, steady-state conditions are assumed, together with uniform thermal conductivity and heat dissipation throughout the panel. Under these assumptions, Equation 1 can be written as:

$$k_{\text{ef, along}} \frac{\partial^2 T}{\partial x^2} + \dot{q_v} = 0, \tag{2}$$

where $k_{\text{ef, along}}$ represents the effective thermal conductivity of the structure, which is modelled as shown in Equation 3:

$$k_{\text{ef, along}} = \sum_{\text{i=1}}^{3} \frac{k_i \delta_i}{\delta_t} = \frac{2\ k_{\text{CFRP}} L_{y\text{CFRP}} + k_{\text{HC}} L_{y\text{HC}}}{\delta_t}, \tag{3}$$

where δ_t represents the total thickness of the panel. For simplicity, the notation k_{ef} will be used from now on instead of $k_{\text{ef, along}}$.

On the other hand, $\dot{q_v}$ represents the internal heat generated by the electronic box per unit volume, which can be expressed as:

$$\dot{q_v} = \frac{W_{\text{EBOX}}}{V}, \tag{4}$$

where V represents the volume of the entire structure ($V = L_{xp} L_{zp} (L_{y\text{HC}} + 2\ L_{y\text{CFRP}})$). The heater is assumed to be switched off. Therefore, after successive integration:

$$\frac{\mathrm{d}^2 T}{\mathrm{d}x^2} = -\frac{\dot{q_v}}{k_{\text{ef}}} \quad \xrightarrow{\int} \quad T(x) = -\frac{\dot{q_v} x^2}{2 k_{\text{ef}}} + \frac{\dot{q_v}}{2 k_{\text{ef}}} L_{xp} x + 273.15, \tag{5}$$

where the following boundary conditions have been imposed: $T(0) = T(L_{xp}) = 273.15$ K.

Secondly, heat dissipation is considered only in the region where both the electronic box and the heater are located. The heat dissipation of the electronic box is distributed uniformly among the panel nodes where it is installed, and the equation describing the steady-state temperature distribution along the spatial coordinate is obtained. In this case, due to the symmetry of the problem, the first half of the spatial domain is divided into three different regions, observed in 4:

- I $\rightarrow 0 < x < x_1$: From the beginning of the panel to the EBOX boundary along the x-axis, with $x_1 = 200$ mm.
- II $\rightarrow x_1 < x < x_2$: From the EBOX boundary to the heater boundary along the x-axis, with $x_2 = 250$ mm.
- III $\rightarrow x_2 < x < x_3$: From the heater boundary to the midpoint of the panel, with $x_3 = 300$ mm.

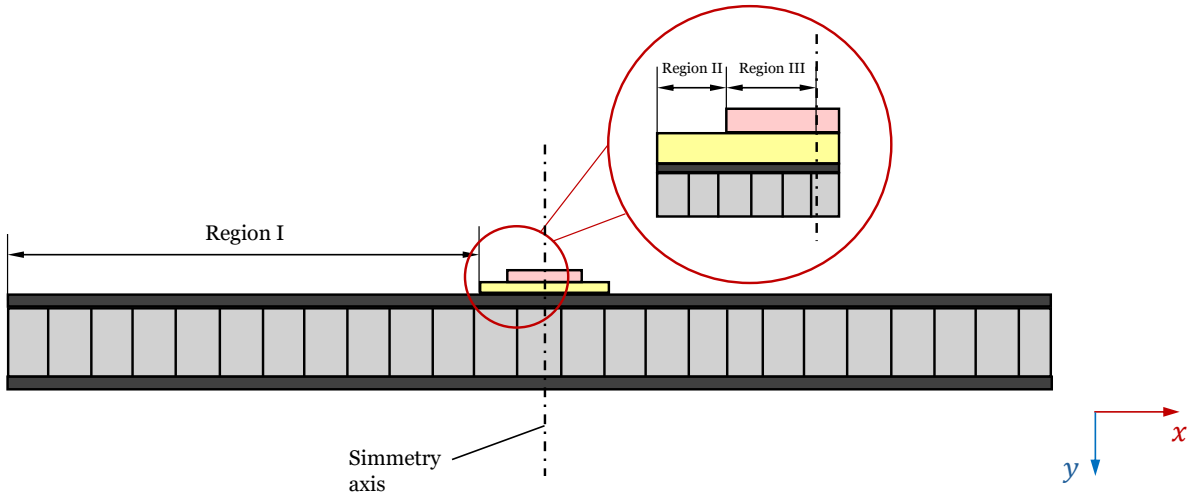


Figure 4. Regions considered for the analytical and numerical model.

By applying successive integration again to Equation 1, using its steady-state form, the following temperature profiles are obtained for each region:

$$T_{\mathrm{I}}(x) = C_1 x + C_2, \tag{6}$$

$$T_{\mathrm{II}}(x) = -\frac{\dot{q_{\mathrm{II}}}}{2k_{\mathrm{ef}}} x^2 + C_3 x + C_4, \tag{7}$$

$$T_{\mathrm{III}}(x) = -\frac{\dot{q_{\mathrm{III}}}}{2k_{\mathrm{ef}}} x^2 + C_5 x + C_6. \tag{8}$$

To determine the integration constants, the following boundary conditions are imposed:

$$T_{\mathrm{I}}(0) = 273.15\ K \tag{9}$$

$$\left.\frac{\partial T_{\mathrm{I}}}{\partial x}\right|_{x_1} = \left.\frac{\partial T_{\mathrm{II}}}{\partial x}\right|_{x_1}, \tag{10}$$

$$T_{\mathrm{I}}(x_1) = T_{\mathrm{II}}(x_1), \tag{11}$$

$$\left.\frac{\partial T_{\mathrm{II}}}{\partial x}\right|_{x_2} = \left.\frac{\partial T_{\mathrm{III}}}{\partial x}\right|_{x_2}, \tag{12}$$

$$T_{\mathrm{II}}(x_2) = T_{\mathrm{III}}(x_2), \tag{13}$$

$$\left.\frac{\partial T_{\mathrm{III}}}{\partial x}\right|_{x_3} = 0, \tag{14}$$

These boundary conditions represent temperature and heat flux continuity at each boundary point (x_1 and x_2), as well as the symmetry condition at the midpoint (x_3). At this point, the heat flux must be zero, which means that the temperature profile must have a zero slope. The integration constants are obtained progressively as follows:

$$C_2 = 273.15 \text{ K}, \tag{15}$$

$$C_5 = \frac{1}{k_{\mathrm{ef}}}(\dot{q_{\mathrm{II}}} + \dot{q_{\mathrm{III}}})x_3, \tag{16}$$

$$C_3 = -\frac{1}{k_{\mathrm{ef}}}\dot{q_{\mathrm{III}}}x_2 + C_5, \tag{17}$$

$$C_1 = -\frac{\dot{q_{\mathrm{II}}}}{k_{\mathrm{ef}}}x_1 + C_3, \tag{18}$$

$$C_4 = \frac{1}{2\ k_{\mathrm{ef}}}\dot{q_{\mathrm{II}}}x_1^2 - C_3x_1 + C_1x_1 + C_2, \tag{19}$$

$$C_6 = (C_3 - C_5)x_2 + C_4, \tag{20}$$

where the heat generation per unit volume in each region is expressed as follows:

$$\dot{q_{\mathrm{II}}} = \frac{W_{\mathrm{EBOX}}}{L_{x\mathrm{EBOX}}L_{zp}\delta_t}, \tag{21}$$

$$\dot{q_{\mathrm{III}}} = \frac{W_{\mathrm{EBOX}}}{L_{x\mathrm{EBOX}}L_{zp}\delta_t} + \frac{W_{\mathrm{HT}}}{L_{x\mathrm{HT}}L_{zp}\delta_t}. \tag{22}$$

2.3. Numerical Modelling

Numerical modelling of Equation (1) is a powerful tool for understanding and analysing the behaviour of complex physical systems involving heat transfer. By discretising this equation, numerical solutions can be obtained, making it possible to study thermal phenomena in greater detail.

Assuming that there are no heat sources or sinks, Equation (1) can be written as:

$$\rho c\frac{\partial T}{\partial t} = k\frac{\partial^2 T}{\partial x^2}. \tag{23}$$

The explicit Euler method is a simple approach in terms of both implementation and understanding. It is also computationally efficient, since it does not require solving a system of equations at each time step. This makes it particularly useful for problems involving a large number of grid points. However, the method has some limitations in terms of stability. Its numerical stability depends on the choice of the time step, which must satisfy certain conditions. If these conditions are not met, the numerical solution may become unstable.

The domain is divided into equally spaced points in both space and time, using a time step Δt and a spatial step Δx, respectively. The subscript i is used to identify the spatial position, while the superscript j denotes the time step.

Using the explicit Euler method, the time derivative can be approximated as:

$$\left.\frac{\mathrm{d}T_i}{\mathrm{d}t}\right|_j \approx \frac{T_i^{j+1} - T_i^j}{\Delta t}. \tag{24}$$

On the other hand, the spatial derivative can be approximated using a central finite difference:

$$k\frac{\partial^2 T}{\partial x^2} \approx k\frac{T_{i-1}^j - 2T_i^j + T_{i+1}^j}{(\Delta x)^2}. \tag{25}$$

Combining the previous expressions, Equation (1) can be written in its discretised form as:

$$\rho c \frac{T_i^{j+1} - T_i^j}{\Delta t} \approx k \frac{T_{i-1}^j - 2T_i^j + T_{i+1}^j}{(\Delta x)^2}, \tag{26}$$

and, rearranging the terms, we obtain:

$$T_i^{j+1} = T_i^j + \frac{k}{\rho c} \frac{1}{(\Delta x)^2} \Delta t \cdot (T_{i-1}^j - 2T_i^j + T_{i+1}^j), \tag{27}$$

which can also be written as:

$$T_i^{j+1} = T_i^j + \alpha \frac{1}{(\Delta x)^2} \Delta t \cdot (T_{i-1}^j - 2T_i^j + T_{i+1}^j), \tag{28}$$

where α is the thermal diffusivity, defined as $\alpha = k/(\rho c)$.

The Fourier number is an important parameter when assessing the stability of a time integration scheme such as the explicit Euler method. It relates the thermal diffusivity of the material to the spatial and temporal steps used in the discretisation. For the explicit Euler method, the numerical stability of the solution depends on this parameter. In the discretised heat equation, the Fourier number, Fo, is defined as:

$$Fo = \frac{(\Delta t)\alpha}{(\Delta x)^2}. \tag{29}$$

To ensure the stability of the explicit Euler scheme, the Fourier number must be less than or equal to a critical value, which is typically 1/2 for the one-dimensional heat equation. If the Fourier number exceeds this limit, the numerical solution may become unstable and produce non-physical results. Therefore, when selecting the time and spatial steps, it is essential to ensure that the Fourier number remains within the stability limit. In this case, the following condition must be satisfied:

$$\Delta t \leq \frac{(\Delta x)^2}{2\alpha}. \tag{30}$$

2.3.1. Steady-State Cases

For the solution of steady-state cases, where the time-dependent term vanishes, the heat equation is simplified and becomes a matrix system to be solved, represented by the expression $[\boldsymbol{A}]\{\boldsymbol{T}\} = \{\boldsymbol{b}\}$, where $[\boldsymbol{A}]$ is a matrix containing the coefficients associated with the temperatures at the panel nodes, and $\{\boldsymbol{T}\}$ is a vector representing the temperature at each panel node. Matrix $[\boldsymbol{A}]$ has a tridiagonal structure, with specific values along the main diagonal and the two adjacent diagonals. Vector $\{\boldsymbol{b}\}$ contains the contributions from heat sources and boundary conditions at each panel node. Matrix $[\boldsymbol{A}]$ is a square matrix with dimensions $N \times N$, where N is the number of nodes in the spatial discretisation. According to the spatial distribution considered, it takes the following form:

$$[\boldsymbol{A}]_{[N\times N]} = \frac{k}{\Delta x^2} \begin{pmatrix} 1 & 0 & 0 & \cdots & \cdots & \cdots & \cdots & 0 \\ 1 & -2 & 1 & 0 & \cdots & \cdots & \cdots & 0 \\ 0 & 1 & -2 & 1 & 0 & \cdots & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 & -2 & 1 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & \cdots & \cdots & 0 & 1 \end{pmatrix},$$

where the steady-state form of Equation (1) has been considered, without including thermal radiation effects or those associated with the contact resistance between the EBOX and the panel. Vector $\{\boldsymbol{T}\}$ can be written as:

$$\{\boldsymbol{T}\} = \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ \vdots \\ T_i \\ \vdots \\ T_{N-1} \\ T_N \end{Bmatrix}, \tag{31}$$

where i takes integer values from 1 to N. On the other hand, vector $\{\boldsymbol{b}\}$ takes the following form:

$$\{\boldsymbol{b}\} = \begin{Bmatrix} T_1 \\ -\frac{\dot{q}_v}{k}(\Delta x)^2 \\ -\frac{\dot{q}_v}{k}(\Delta x)^2 \\ \vdots \\ -\frac{\dot{q}_v}{k}(\Delta x)^2 \\ \vdots \\ -\frac{\dot{q}_v}{k}(\Delta x)^2 \\ T_N \end{Bmatrix}, \tag{32}$$

where the boundary conditions have been applied to the first and last components of the vector. A uniform thermal conductivity and heat dissipation throughout the panel have been assumed, with the EBOX again considered to extend along the entire shorter side of the panel.

If the heat dissipation from the EBOX and the heater is positioned as shown in Figure (**??**), the spatial domain is divided into different regions along the x-direction, as described in the section dealing with the analytical solution. In this case, the changes are reflected only in vector $\{\boldsymbol{b}\}$. Some of its components are zero, corresponding to Region I; others account exclusively for the heat dissipation from the EBOX, corresponding to Region II; and the remaining components correspond to the region between the EBOX and the heater, where the contributions from both are considered if the heater is switched on, corresponding to Region III.

To ensure a convenient spatial discretisation for the numerical study, the domain is divided in such a way that there is always a node located at x_1, x_2 and x_3. This means that the total number of nodes in the spatial discretisation must satisfy the following condition:

$$N = 7 + 6 \cdot (2 \cdot n - 1), \tag{33}$$

where n can only take positive integer values different from zero.

2.3.2. EBOX as an Independent Node

The EBOX is considered as a separate node from the panel, being thermally connected to the panel nodes through the contact conductance specified in Table (2). Furthermore, both the thermal dissipation and the thermal capacity of the EBOX are assumed to be concentrated at this specific node.

The numerical equation (for the steady-state case) for the nodes belonging to the panel is expressed as:

$$T_{i-1} - 2T_i + T_{i+1} + \frac{(\Delta x)^2}{k} h A_{ci} \frac{T_{\text{EBOX}} - T_i}{\Delta V_i} = 0, \tag{34}$$

where the heater is assumed to be switched off. The variable A_{ci} denotes the contact area between a cell and the EBOX for the corresponding nodes. Therefore, for node i:

$$A_{ci} = L_{z\text{EBOX}} \Delta x. \tag{35}$$

The variable h denotes the contact conductance between the panel and the EBOX. The variable ΔV_i represents the volume of a cell in the spatial discretisation. For an intermediate node not located at x_2, it is given by:

$$\Delta V_i = L_{zp} \Delta x \delta_t, \tag{36}$$

where δ_t represents the total thickness of the panel. Both the definition of ΔV_i and that of A_{ci} assume that node i is not located at x_2. If it coincides with x_2, a factor of $1/2$ must be included to account only for the half of the cell area that is in contact with the EBOX. Substituting Equation (36) into Equation (34), we obtain:

$$T_{i-1} + T_{i+1} - \left(2 + \frac{(\Delta x)^2}{k} h_{ef} \frac{1}{\delta_t}\right) \cdot T_i + \frac{(\Delta x)^2}{k} h_{ef} \frac{1}{\delta_t} T_{\text{EBOX}} = 0, \tag{37}$$

where h_{ef} denotes the effective contact conductance between the panel and the EBOX, defined as:

$$h_{\text{ef}} = \frac{L_{z\text{EBOX}}}{L_{zp}} h. \tag{38}$$

On the other hand, an equation is established to describe the thermal balance of the EBOX. The heat generated internally is dissipated into the panel through the thermal resistances connecting the EBOX to the panel nodes in contact with it. This can be expressed mathematically as:

$$W_{\text{EBOX}} = \sum_{i=1}^{N_e} h A_{ci} (T_{\text{EBOX}} - T_i), \tag{39}$$

where N_e represents the number of panel nodes in contact with the EBOX. This equation can be rewritten as:

$$\Phi = (N_e - 1) T_{\text{EBOX}} - \sum_{i=1}^{N_e} T_i = \frac{W_{\text{EBOX}}}{h \Delta x L_{z\text{EBOX}}}. \tag{40}$$

Considering the steady-state case once again, the problem can be solved by determining the solution of the system $[\boldsymbol{A}]\{\boldsymbol{T}\} = \{\boldsymbol{b}\}$. The elements of matrix $[\boldsymbol{A}]$ are modified along its three diagonals according to the formulation presented in Equation 36. For an intermediate node i in contact with the EBOX, but not coinciding with its boundary at x_2, the tridiagonal coefficients take the following form, for $(N - N_e + 2)/2 < i < N_e$:

$$[a_{i,i-1} \;\; a_{i,i} \;\; a_{i,i+1}] = [1 \;\; -2 - h_{\text{ef}} (\Delta x)^2 \frac{1}{k \delta_t} \;\; 1]. \tag{41}$$

On the other hand, the two nodes located at the boundaries of the EBOX have the following tridiagonal coefficients:

$$[a_{i,i-1} \;\; a_{i,i} \;\; a_{i,i+1}] = [1 \;\; -2 - \frac{1}{2} h_{\text{ef}} (\Delta x)^2 \frac{1}{k \delta_t} \;\; 1], \tag{42}$$

where the fact that the contact area with the EBOX is half that of a complete cell has been taken into account.

Let the first node in contact with the EBOX be node N_p and the last one be node N_q, such that $N_q - N_p = N_e - 1$. Since there are N nodes in the panel and one additional node representing the EBOX, the last column of matrix $[\boldsymbol{A}]$ takes the following form:

$$[\boldsymbol{A}]_{[:,N+1]} = \begin{Bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \frac{1}{2} h_{\text{ef}} (\Delta x)^2 \frac{1}{k \delta_t} \\ h_{\text{ef}} (\Delta x)^2 \frac{1}{k \delta_t} \\ \vdots \\ h_{\text{ef}} (\Delta x)^2 \frac{1}{k \delta_t} \\ \frac{1}{2} h_{\text{ef}} (\Delta x)^2 \frac{1}{k \delta_t} \\ 0 \\ \vdots \\ 1 \\ N_q - N_p \end{Bmatrix}, \tag{43}$$

On the other hand, according to Equation (39), the last row of matrix $[\boldsymbol{A}]$ takes the following form:

$$[\boldsymbol{A}]_{[N+1,:]} = \left\{ 0 \;\; \cdots \;\; 0 \;\; -\frac{1}{2} \;\; -1 \;\; -1 \;\; \cdots \;\; -1 \;\; -\frac{1}{2} \;\; 0 \;\; \cdots \;\; 0 \;\; N_q - N_p \right\} \tag{44}$$

Finally, under these conditions, vector $\{\boldsymbol{b}\}$ can be expressed as follows, according to Equation (40):

$$\{\boldsymbol{b}\} = \begin{Bmatrix} T_1 \\ 0 \\ \vdots \\ 0 \\ \\ \vdots \\ 0 \\ \vdots \\ 0 \\ T_N \\ \Phi \end{Bmatrix}. \tag{45}$$

2.3.3. Complete Transient Modelling

For the complete transient modelling, an explicit Euler time integration scheme is used to solve the following equation:

$$\{\boldsymbol{T^{j+1}}\} = \{\boldsymbol{T^{j}}\} + \frac{\Delta t}{\bar{\rho} \cdot c_{\text{ef}} \cdot \Delta V_i}\left(\frac{k}{(\Delta x)^2}\Delta V_i\{\boldsymbol{\phi_s}\} + \{\boldsymbol{\phi_c}\} + \{\boldsymbol{\phi_r}\} + \{\boldsymbol{\phi_{HT}}\}\right) + \frac{\Delta t}{c_{\text{EBOX}}}\left(\{\boldsymbol{\psi}_{\text{EBOX}}\} + \{\boldsymbol{\psi_c}\} + \{\boldsymbol{\psi_r}\}\right), \tag{46}$$

where the equation has been divided into two main terms: the first accounts for the effects acting on the panel nodes, while the second accounts for the effects acting directly on the EBOX.

First, both the density and the specific heat capacity of the system have been averaged. Thus, $\bar{\rho}$ and c_{ef} are defined as:

$$\bar{\rho} = \frac{\rho_{\text{HC}} L_{y_{\text{HC}}} + 2\rho_{\text{CFRP}} L_{y_{\text{CFRP}}}}{\delta_t}, \tag{47}$$

$$c_{\text{ef}} = \frac{2 \cdot \rho_{\text{CFRP}} \cdot c_{\text{CFRP}} \cdot V_{\text{CFRP}} + \rho_{\text{HC}} \cdot c_{\text{HC}} \cdot V_{\text{HC}} + c_{\text{EBOX}}}{V_t \cdot \bar{\rho}}, \tag{48}$$

where $V_{\text{CFRP}} = L_{xp} L_{zp} L_{y_{\text{CFRP}}}$ and $V_{\text{HC}} = L_{xp} L_{zp} L_{y_{\text{HC}}}$ represent the volumes of the CFRP and Honeycomb layers, respectively, and V_t is the total volume of the panel. The thermal capacity of the EBOX has also been included, following the approach described in [2].

The term $\{\boldsymbol{\phi_s}\}$ represents the second spatial derivative. Taking into account that, in Equation (46), the last node (and therefore the last component of the different vectors) corresponds to the EBOX, $\{\boldsymbol{\phi_s}\}$ can be written as:

$$\{\boldsymbol{\phi_s}\} = \begin{Bmatrix} 0 \\ T_1 - 2 \cdot T_2 + T_3 \\ T_2 - 2 \cdot T_3 + T_4 \\ \vdots \\ \\ T_{i-1} - 2 \cdot T_i + T_{i+1} \\ \vdots \\ T_{N-2} - 2 \cdot T_{N-1} + T_N \\ 0 \\ 0 \end{Bmatrix}. \tag{49}$$

The term $\{\boldsymbol{\phi_c}\}$ accounts for heat conduction through the contact interface between the panel and the EBOX, and is

given by:

$$\{\boldsymbol{\phi_c}\} = \begin{Bmatrix} 0 \\ \vdots \\ hL_{z\mathrm{EBOX}}\dfrac{\Delta x}{2}(T_{\mathrm{EBOX}} - T_{N_p}) \\ hL_{z\mathrm{EBOX}}\Delta x(T_{\mathrm{EBOX}} - T_{N_p+1}) \\ \vdots \\ hL_{z\mathrm{EBOX}}\Delta x(T_{\mathrm{EBOX}} - T_{N_q-1}) \\ hL_{z\mathrm{EBOX}}\dfrac{\Delta x}{2}(T_{\mathrm{EBOX}} - T_{N_q}) \\ 0 \\ \vdots \\ 0 \end{Bmatrix}. \tag{50}$$

The third term accounts for thermal radiation. It is assumed that the irradiance on the lateral face of the panel has a negligible influence compared with that on its upper and lower faces. It is also assumed that the EBOX extends along the entire shorter side of the panel and that the heater does not emit thermal radiation. Under these assumptions, the term $\{\boldsymbol{\phi_r}\}$ is expressed as:

$$\{\boldsymbol{\phi_r}\} = \begin{Bmatrix} \sigma\varepsilon\Delta x L_{zp}\cdot(T_R^4 - T_1^4) \\ \sigma\varepsilon 2\Delta x L_{zp}\cdot(T_R^4 - T_2^4) \\ \vdots \\ \sigma\varepsilon\dfrac{3}{2}\Delta x L_{zp}\cdot(T_R^4 - T_{N_p}^4) \\ \sigma\varepsilon L_{zp}\Delta x\cdot(T_R^4 - T_{N_p+1}^4) \\ \vdots \\ \sigma\varepsilon L_{zp}\Delta x\cdot(T_R^4 - T_{N_q-1}^4) \\ \sigma\varepsilon L_{zp}\dfrac{3}{2}\Delta x\cdot(T_R^4 - T_{N_q}^4) \\ \sigma\varepsilon 2\Delta x L_{zp}\cdot(T_R^4 - T_2^4) \\ \vdots \\ \sigma\varepsilon\Delta x L_{zp}\cdot(T_R^4 - T_N^4) \\ 0 \end{Bmatrix}, \tag{51}$$

where σ is the Stefan–Boltzmann constant ($\sigma = 5.67\cdot 10^{-8}$), and ε is the emissivity of the CFRP and the EBOX, which is assumed to be 0.8. The temperature T_R represents the temperature of the satellite, which is modelled as a black body at 0°C.

For the heater, when it is switched on, the corresponding term is:

$$\{\boldsymbol{\phi_{HT}}\} = \begin{Bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \dfrac{1}{2}\dfrac{W_{\mathrm{HT}}}{L_{x_{\mathrm{HT}}}L_{z_{\mathrm{HT}}}\delta_t}\Delta x\delta_t L_{zp} \\ \dfrac{W_{\mathrm{HT}}}{L_{x_{\mathrm{HT}}}L_{z_{\mathrm{HT}}}\delta_t}\Delta x\delta_t L_{zp} \\ \vdots \\ \dfrac{W_{\mathrm{HT}}}{L_{x_{\mathrm{HT}}}L_{z_{\mathrm{HT}}}\delta_t}\Delta x\delta_t L_{zp} \\ \dfrac{1}{2}\dfrac{W_{\mathrm{HT}}}{L_{x_{\mathrm{HT}}}L_{z_{\mathrm{HT}}}\delta_t}\Delta x\delta_t L_{zp} \\ 0 \\ \vdots \\ 0 \end{Bmatrix}, \tag{52}$$

where the non-zero components correspond to the panel nodes in contact with the heater.

Regarding the EBOX, its internal heat dissipation is represented by:

$$\{\boldsymbol{\psi}_{\text{EBOX}}\} = \begin{Bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ W_{\text{EBOX}} \end{Bmatrix}, \tag{53}$$

while the heat conduction term from the panel to the EBOX is given by:

$$\{\boldsymbol{\psi_c}\} = \begin{Bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \sum_{i=N_p+1}^{N_q-1} hL_{z\text{EBOX}}\Delta x(T_i - T_{\text{EBOX}}) + \frac{1}{2}hL_{z\text{EBOX}}\Delta x(T_{N_p} - T_{\text{EBOX}}) + \frac{1}{2}hL_{z\text{EBOX}}\Delta x(T_{N_q} - T_{\text{EBOX}}) \end{Bmatrix}, \tag{54}$$

where heat transfer from the panel to the EBOX is considered.

Finally, the contribution of thermal radiation to the EBOX is expressed as:

$$\{\boldsymbol{\psi_r}\} = \begin{Bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \sigma\varepsilon L_{x_{\text{EBOX}}} L_{z_{\text{EBOX}}} (T_R^4 - T_{\text{EBOX}}^4) \end{Bmatrix}. \tag{55}$$

In Equation (46), the problem is solved starting from the initial temperature conditions, $\{\boldsymbol{T^j}\}$. For the case considered in this study, all nodes are initially set to a temperature of 5°C, except for the nodes located at the ends of the panel (1 and $N-1$), which are maintained at 0°C at all times:

$$\{\boldsymbol{T^j}\} = \begin{Bmatrix} 273.15 \\ 278.15 \\ \vdots \\ 278.15 \\ \vdots \\ 278.15 \\ 273.15 \\ 278.15 \end{Bmatrix}. \tag{56}$$

Again, it is important to remember that the last raw and column of the mathematical system compute the thermal equation for the EBOX, and thus, the boundary conditions as well for the last component of $\boldsymbol{T^j}$ and $\boldsymbol{b}$ vectors.

2.3.4. Heater Control

Starting from the previous steady-state condition, it is assumed that the EBOX switches off due to a failure. To implement thermostat control using the heater, so that the EBOX temperature remains within the range of 5°C to 10°C, an `if` statement is introduced in the code. This ensures that Equation (52) only takes non-zero values when the EBOX temperature falls below its minimum survival temperature, and remains active while the temperature rises until it reaches 10°C. At this point, the term becomes zero again, causing the EBOX temperature to decrease until the heater is activated once more. This process establishes a cycle that produces a sawtooth-shaped temperature profile over time.

Suppose that the EBOX temperature is to be controlled using a Proportional-Integral (PI) controller to maintain it within a desired range, for example, between 5°C and 10°C. The desired temperature is denoted by T_d. The PI controller adjusts the heater power based on the error between the desired temperature and the current EBOX temperature at each time step, T_{EBOX}^j. The current error, E^j, is therefore defined as:

$$E^j = T_d - T_{\text{EBOX}}^j. \tag{57}$$

The control signal, U, generated by the PI controller is calculated as the sum of the proportional term, U_p, and the integral term, U_I:

$$U^j = U_p^j + U_I^j. \tag{58}$$

The proportional term, U_p, is calculated by multiplying the error by a proportional gain, K_p:

$$U_p^j = K_p \cdot E^j. \tag{59}$$

The integral term, U_I, is calculated by integrating the error over time and multiplying it by an integral gain, K_I:

$$U_I^j = K_I \int_0^{t^j} E(t)\,\mathrm{d}t. \tag{60}$$

However, if the integral control is allowed to act immediately after the EBOX and the heater are switched off, the accumulated error would increase rapidly due to the large initial error. This could lead to an undesirable system response, such as the heater being activated at maximum power too quickly, potentially causing temperature oscillations or instability. Therefore, the integral control is not allowed to act immediately after the EBOX and the heater are switched off. Instead, the proportional control is expected to provide most of the initial adjustment to the heater power in order to reduce the initial error. Once the system has stabilised and the accumulated error has reached a manageable level, the integral control is enabled after a specified time to correct small deviations of the EBOX temperature from the desired value.

3. Results

This section presents the results obtained by applying the analytical and numerical methods discussed in the previous sections. The different parts of the study are examined in detail, showing the solutions obtained for each case and highlighting their main characteristics and their impact on understanding the thermal behaviour of the system under study. The results are presented in an organised and structured manner, emphasising the main conclusions drawn from each analysis. In addition, the analytical and numerical solutions are compared in order to validate the models used and assess their accuracy and reliability.

3.1. First Case

In the first case, a uniform distribution of both thermal conductivity and heat dissipation throughout the entire panel is assumed. This approach makes it possible to analyse the thermal behaviour of the system under simplified conditions, where all regions of the panel are considered to have uniform thermal properties. The main objective of this case is to evaluate how the temperature distribution across the panel changes under these uniform conditions, providing an initial basis for comparison and a better understanding of the effects of more complex configurations considered in the following cases.

Figure 5 illustrates the influence of the number of nodes on the accuracy of the numerical temperature approximation. In addition, a certain degree of symmetry can be observed in the results, as expected, since the problem is symmetric in terms of both its dimensional and thermal properties.

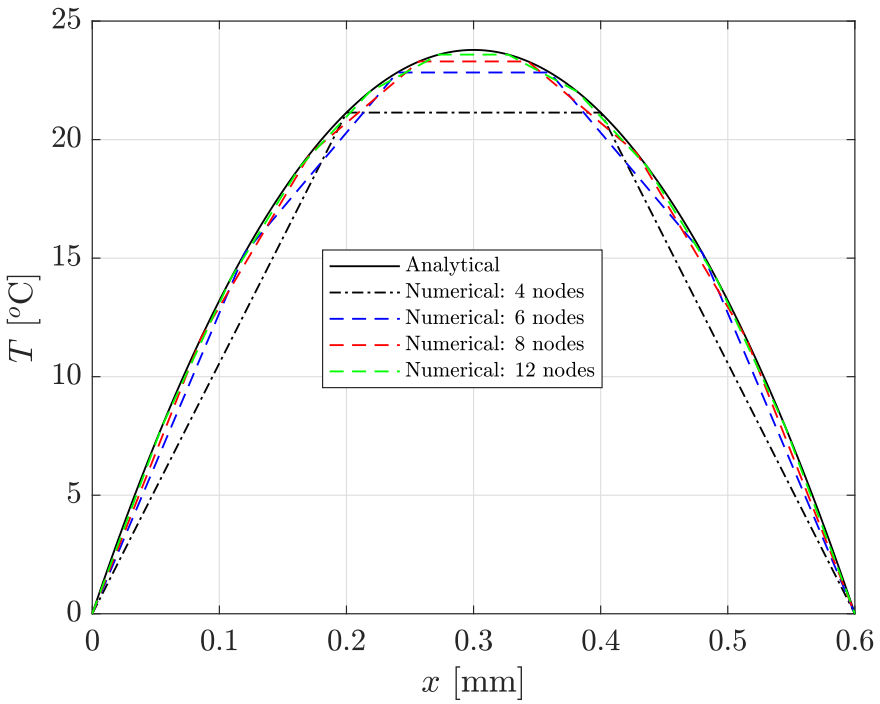


Figure 5. Comparison between the analytical and numerical results for different spatial discretisations, assuming a uniform distribution of both thermal conductivity and heat dissipation throughout the entire panel.

A gradual increase in temperature can be observed as the position x moves away from the ends of the panel, where a zero-temperature boundary condition has been imposed. This increase reaches its maximum at the midpoint of the structure, where the heat flux is zero and the temperature reaches its highest value. The maximum temperature reached in the system is 296.93 K.

3.2. Second Case

In the second case, heat dissipation is concentrated in the regions where the EBOX and the heater are located. This approach aims to examine the effect of concentrating heat dissipation in specific areas and to better understand how these local variations influence the thermal behaviour of the system.

The symmetry of the satellite is taken into account in this case by imposing the symmetry boundary condition (Equation 14). The results obtained using both the analytical and numerical approaches are shown in Figure 6. It can be observed that, since heat dissipation is concentrated in a smaller and more localised region compared with the case shown in Figure 5, the maximum temperature reached is higher. This is because the heat generated by the EBOX is transferred to a smaller surface area, resulting in a greater amount of heat absorbed per unit area. In fact, this concentration of heat can lead to a maximum temperature approximately twice as high as that obtained in the previous case. The maximum temperature reached in the system is 314.76 K.

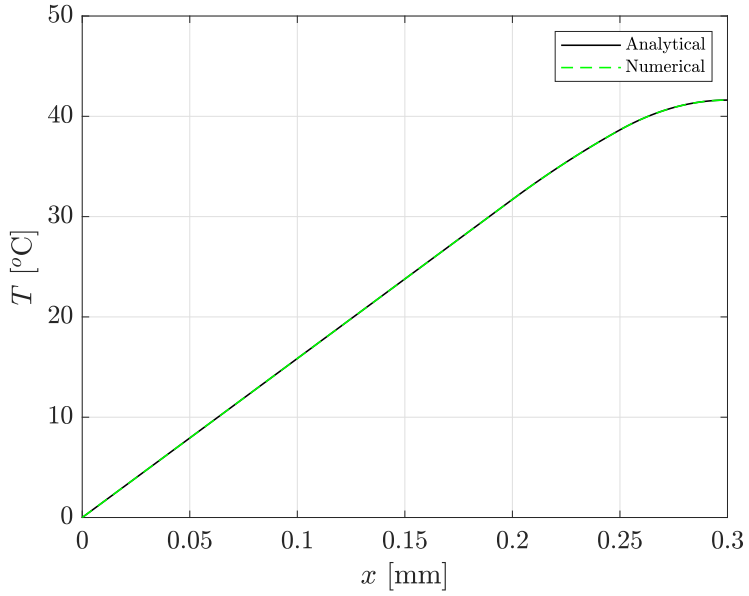


Figure 6. Comparison between the analytical and numerical results using 2000 nodes, assuming a uniform thermal conductivity and heat dissipation concentrated in the region occupied by the EBOX (half of the structure).

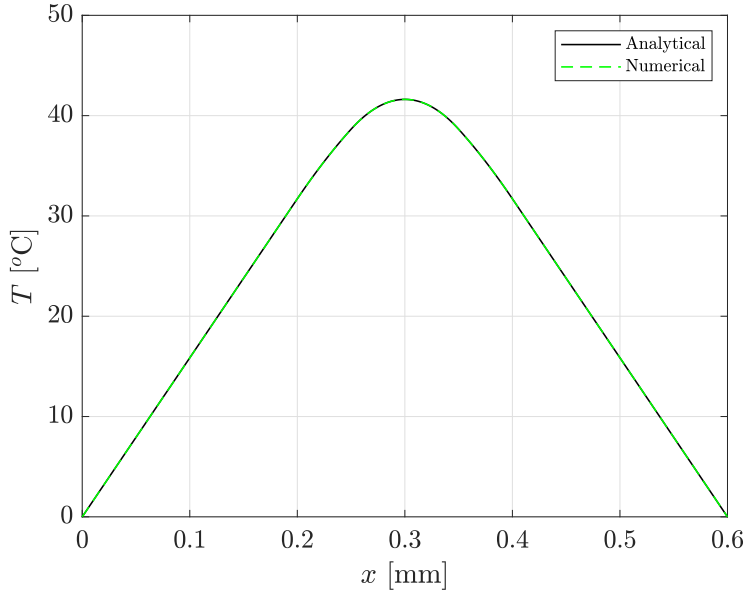


Figure 7. Comparison between the analytical and numerical results using 2000 nodes, assuming a uniform thermal conductivity and heat dissipation concentrated in the region occupied by the EBOX.

3.3. Third Case

In the third case, the EBOX is modelled as a separate node from the satellite structural panel. The EBOX is thermally connected to the panel nodes through the contact conductance specified in Table 2. In addition, both the heat dissipation and the thermal capacity of the EBOX are concentrated in this specific node. The main objective of this case is to calculate the new steady-state temperature profile across the panel, taking into account this configuration in which the EBOX is treated as an independent element. This makes it possible to evaluate how modelling the EBOX as a separate node affects the temperature distribution across the panel and to gain a better understanding of its influence on the thermal behaviour of the system.

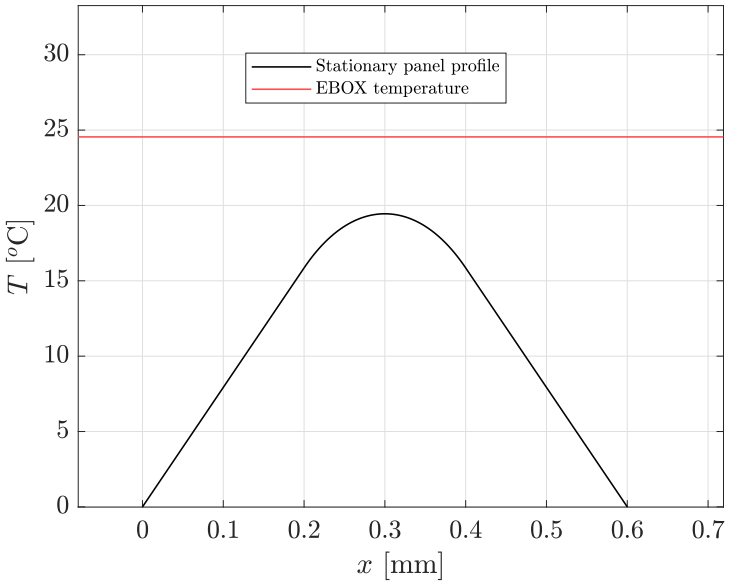


Figure 8. Comparison between the analytical and numerical results using 101 nodes, assuming a uniform thermal conductivity and heat dissipation concentrated in the region occupied by the EBOX, which is modelled as an independent node.

The steady-state temperature profile of the panel under these conditions is shown in Figure 8. It can be observed that the maximum temperature reached is lower as a result of the contact resistance introduced in this case. The maximum temperature reached by the panel is 292.59 K, while the EBOX reaches a maximum temperature of 297.70 K.

3.4. Fourth Case

In the fourth case, heat losses due to thermal radiation towards the interior of the satellite are included. The satellite interior is modelled as a black body at 0°C. An emissivity of 0.8 is assumed for both the CFRP and the EBOX.

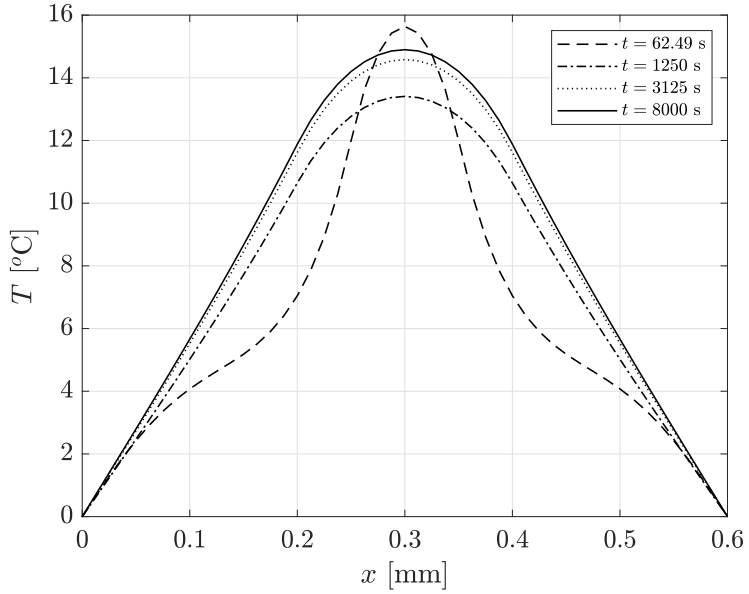


Figure 9. Comparison of the temperature profiles of the different panel nodes at different time instants. The steady-state temperature profile is approximately reached after 8000 s.

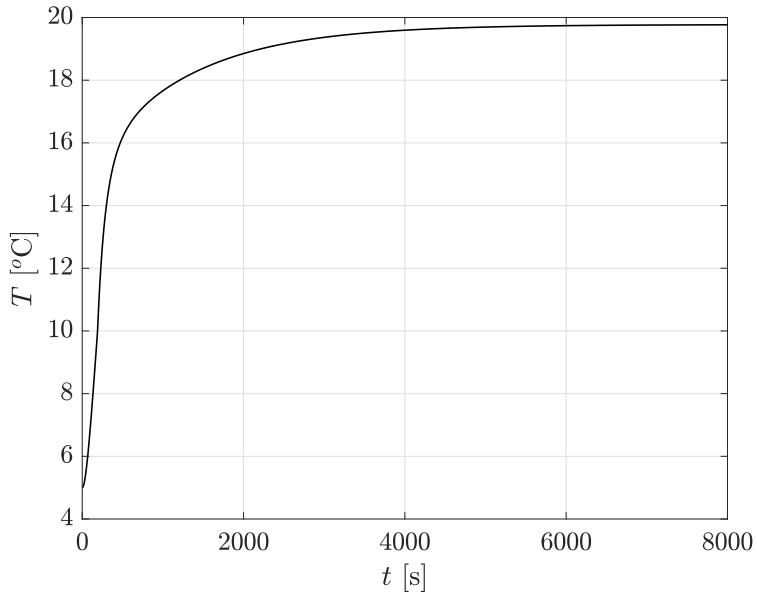


Figure 10. Time evolution of the EBOX temperature until it reaches steady-state conditions.

A transient numerical scheme is used to simulate the thermal behaviour of the system. The main objective of this case is to simulate the heater activation process, starting from an initial condition in which both the panel and the EBOX are switched off and at the minimum survival temperature of 5°C.

The aim is to reach a temperature equal to or higher than the EBOX turn-on requirement. Once this temperature is reached, the EBOX is switched on and the heater is switched off. The time required for the system to reach steady-state conditions is then determined, and the corresponding temperature profile is analysed. This analysis provides insight into how the different thermal components of the system interact during the activation and shutdown processes.

The steady-state temperature profile is shown in Figure 9. A significant initial temperature increase can be observed in the region around the EBOX, followed by the gradual dissipation of heat through the panel towards its edges over time. This process initially leads to a progressive decrease in the maximum temperature. Subsequently, the temperature increases again until steady-state conditions are reached, where temperature variations over time become negligible. Under these steady-state conditions, the maximum panel temperature is 288.047 K. Meanwhile, the EBOX temperature gradually increases over time until steady state is reached, stabilising at 292.92 K, as shown in Figure 10.

3.5. Fifth Case

In the fifth case, heat losses due to thermal radiation towards the interior of the satellite are once again considered. The satellite interior is modelled as a black body at 0°C, with an emissivity of 0.8 for both the CFRP and the EBOX. A transient numerical scheme is used to simulate the time evolution of the system temperature. The main objective of this case is to analyse the evolution of the EBOX temperature following a failure that causes it to shut down, starting from the previously reached steady-state condition. In addition, a thermostat control strategy using the heater is proposed to keep the EBOX temperature within the desired range of 5°C to 10°C.

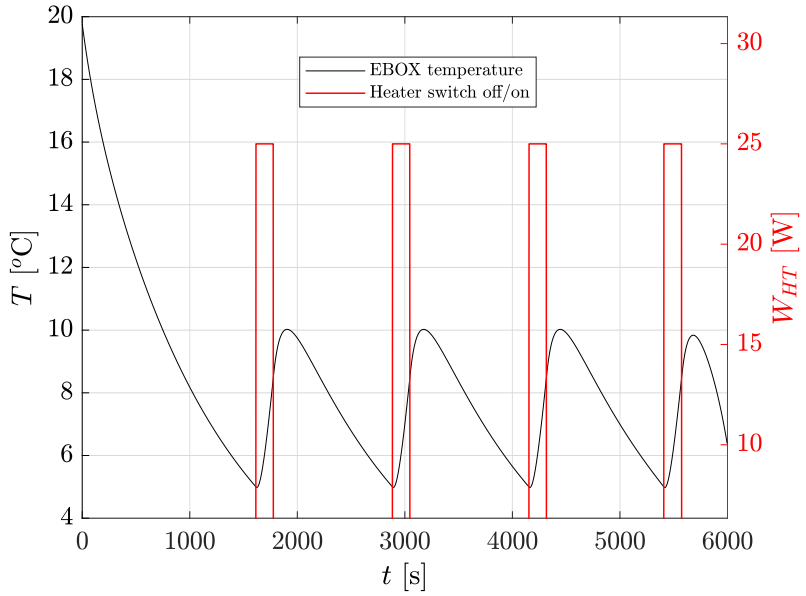


Figure 11. Time evolution of the EBOX temperature between 5°C and 10°C, achieved by controlling the heater switching cycles.

Figure 11 shows the time evolution of the EBOX temperature as a result of the heater switching cycles, which are designed to keep the EBOX temperature within the target range of 5°C to 10°C. The temperature increase and decrease in response to these events can be clearly observed. The time intervals during which the heater is activated are also indicated. The heater is switched on when the EBOX temperature falls below the minimum temperature threshold, causing the EBOX temperature to increase. However, the heater is not switched off when the temperature reaches 10°C, but rather at a lower temperature. This is because, if the heater were switched off only when the EBOX temperature reached 10°C, the temperature would exceed this value due to the residual heat remaining in the vicinity of the EBOX as a result of the previous heating process.

3.6. Sixth Case

In the sixth case, the situation described in the previous case is considered once again, but this time a Proportional-Integral (PI) controller is implemented instead of thermostat control for the heater. The main objective of this case is to analyse how the use of a PI controller affects the temperature regulation of the EBOX. The PI controller uses both the current error and the accumulated error over time to adjust the power supplied to the heater and keep the EBOX temperature within the desired range.

A controller is designed to adjust the heater power according to the error between the current EBOX temperature and the desired temperature. The proportional term adjusts the heater power in proportion to the current error, while the integral term accounts for the accumulated error over time. However, it is essential to prevent the integral control from having an immediate effect after the EBOX and the heater are switched off, since the initial error could be significant and lead to an unstable system response. Instead, the proportional control is allowed to act first, providing most of the initial adjustment to the heater power. Only after the system has stabilised is the integral control enabled to correct smaller deviations from the desired temperature.

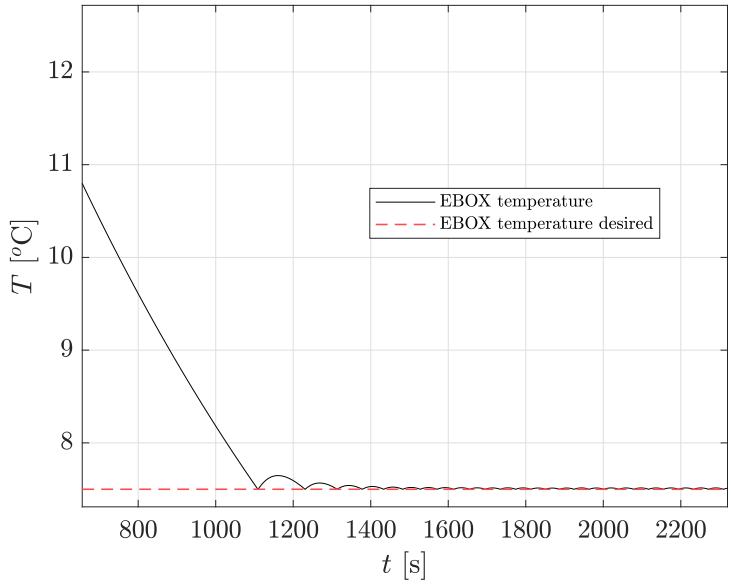


Figure 12. Time evolution of the EBOX temperature following the implementation of the Proportional-Integral Control Law, with a desired temperature of 7.5°C.

4. Conclusions

A progressive analysis has been carried out, ranging from initial simplified models to more detailed and complex approaches. First, a basic one-dimensional model was developed, assuming a uniform distribution of thermal conductivity and heat dissipation throughout the entire panel. This approach made it possible to understand the general effects of heat conduction within the system and establish a solid foundation for further investigations.

Subsequently, the model was refined by considering the heat dissipation of the EBOX exclusively within the region it occupies and, later, by modelling it as a separate node from the panel, with a specific thermal contact conductance between them. This approach allowed the temperature distribution across the panel to be studied more accurately, taking into account the direct influence of the EBOX heat dissipation on the thermal profile. It was concluded that concentrating the EBOX heat dissipation in a smaller region increases the maximum temperature peak of the panel. This effect contrasts with that of the contact resistance, which reduces the heat flow and consequently lowers the maximum temperature peak.

The model was further refined by incorporating heat losses due to thermal radiation towards the interior of the satellite, modelled as a black body at 0°C, and by adopting a transient numerical scheme. This step made it possible to capture the transient and dynamic behaviour of the system more accurately, as well as to evaluate the impact of the initial conditions on the EBOX temperature.

Different control strategies were also explored, including thermostat control and a Proportional-Integral (PI) Control Law, to keep the EBOX temperature within acceptable limits. The results demonstrate that a PI controller can effectively regulate the temperature within the desired range, avoiding oscillations and ensuring stable system behaviour around the target temperature.

This step-by-step approach, progressing from simplified models to more complex and detailed configurations, has made it possible to identify and progressively address the different aspects and challenges associated with the design and operation of the thermal system under study.

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